

TE Curve Verification Pipeline Familywise ONNX Report - wave4_1_log_cosh_robust_loss

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_log_cosh_robust_loss/2026-07-13-11-52-34__track2_wave4_1_log_cosh_robust_loss_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_log_cosh_robust_loss/2026-07-13-11-52-34__track2_wave4_1_log_cosh_robust_loss_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_log_cosh_robust_loss/2026-07-13-11-52-34__track2_wave4_1_log_cosh_robust_loss_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_log_cosh_robust_loss/2026-07-13-11-52-34__track2_wave4_1_log_cosh_robust_loss_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_1_log_cosh_robust_loss` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_1_log_cosh_robust_loss_fw_simplified_setpoints</code>	<code>2026-07-13-03-10-51_te_wave4_1_log_cosh_robust_loss_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_1_log_cosh_robust_loss_bw_simplified_setpoints</code>	<code>2026-07-13-04-12-33_te_wave4_1_log_cosh_robust_loss_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_1_log_cosh_robust_loss_global_simplified_setpoints</code>	<code>2026-07-13-02-40-42_te_wave4_1_log_cosh_robust_loss_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/forward/2026-07-13-03-10-51_te_wave4_1_log_cosh_robust_loss_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/forward/2026-07-13-03-10-51_te_wave4_1_log_cosh_robust_loss_fw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=101-val_mae=0.00354228.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/backward/2026-07-13-04-12-33_te_wave4_1_log_cosh_robust_loss_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/backward/2026-07-13-04-12-33_te_wave4_1_log_cosh_robust_loss_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=099-val_mae=0.00352715.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/global/2026-07-13-02-40-42_te_wave4_1_log_cosh_robust_loss_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/global/2026-07-13-02-40-42_te_wave4_1_log_cosh_robust_loss_global_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=120-val_mae=0.00357307.ckpt</code>

Aggregate Metrics

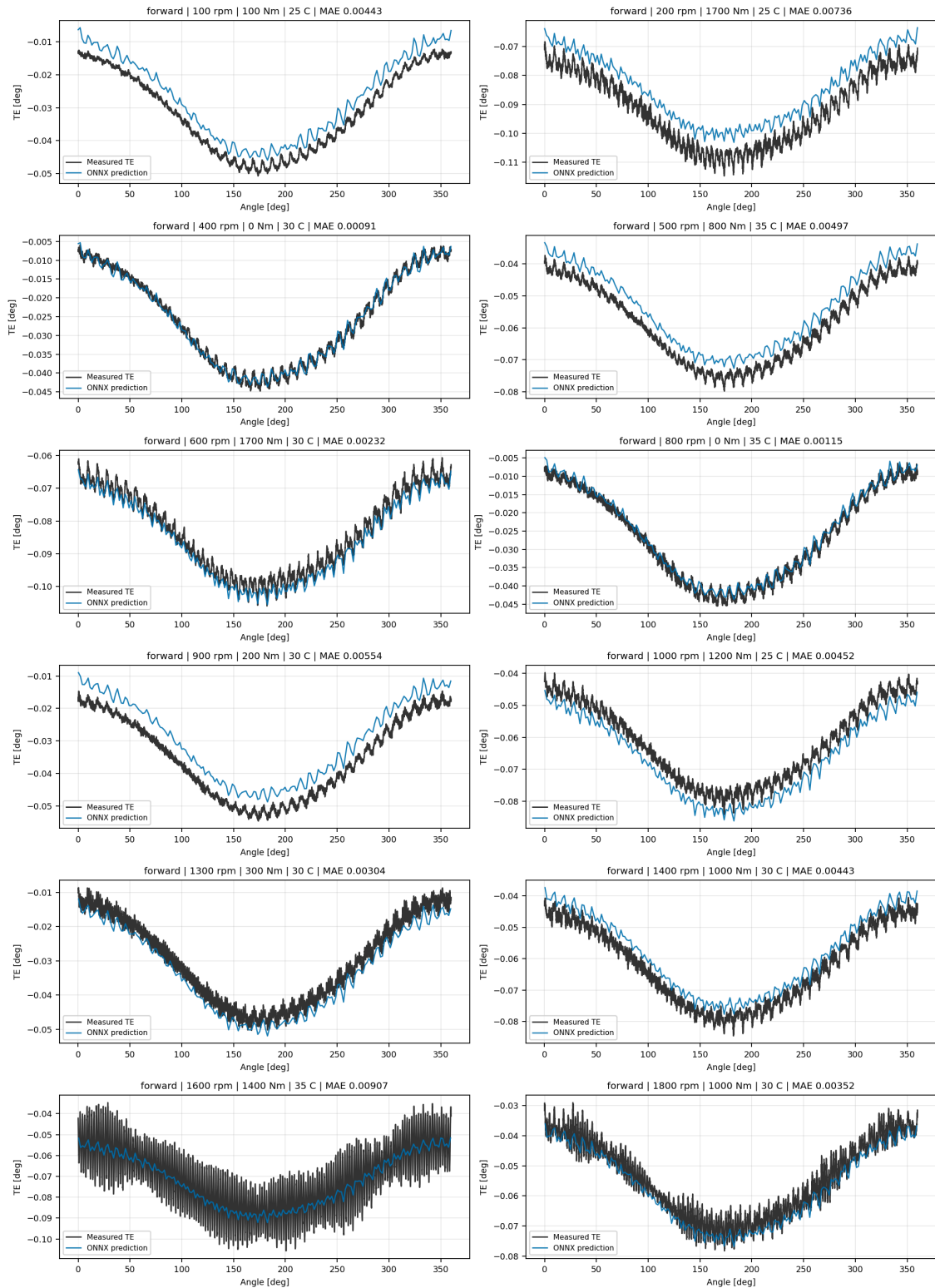
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003296	0.003584	7.713	13.598
backward	97	0.003455	0.003773	8.032	15.847
global	194	0.003339	0.003651	7.785	13.889

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000413	0.002894	0.001302	0.002898
backward	0.000821	0.002794	0.001534	0.003547
global	-0.000001	0.002807	0.001447	0.003166

Forward 12-Curve Page

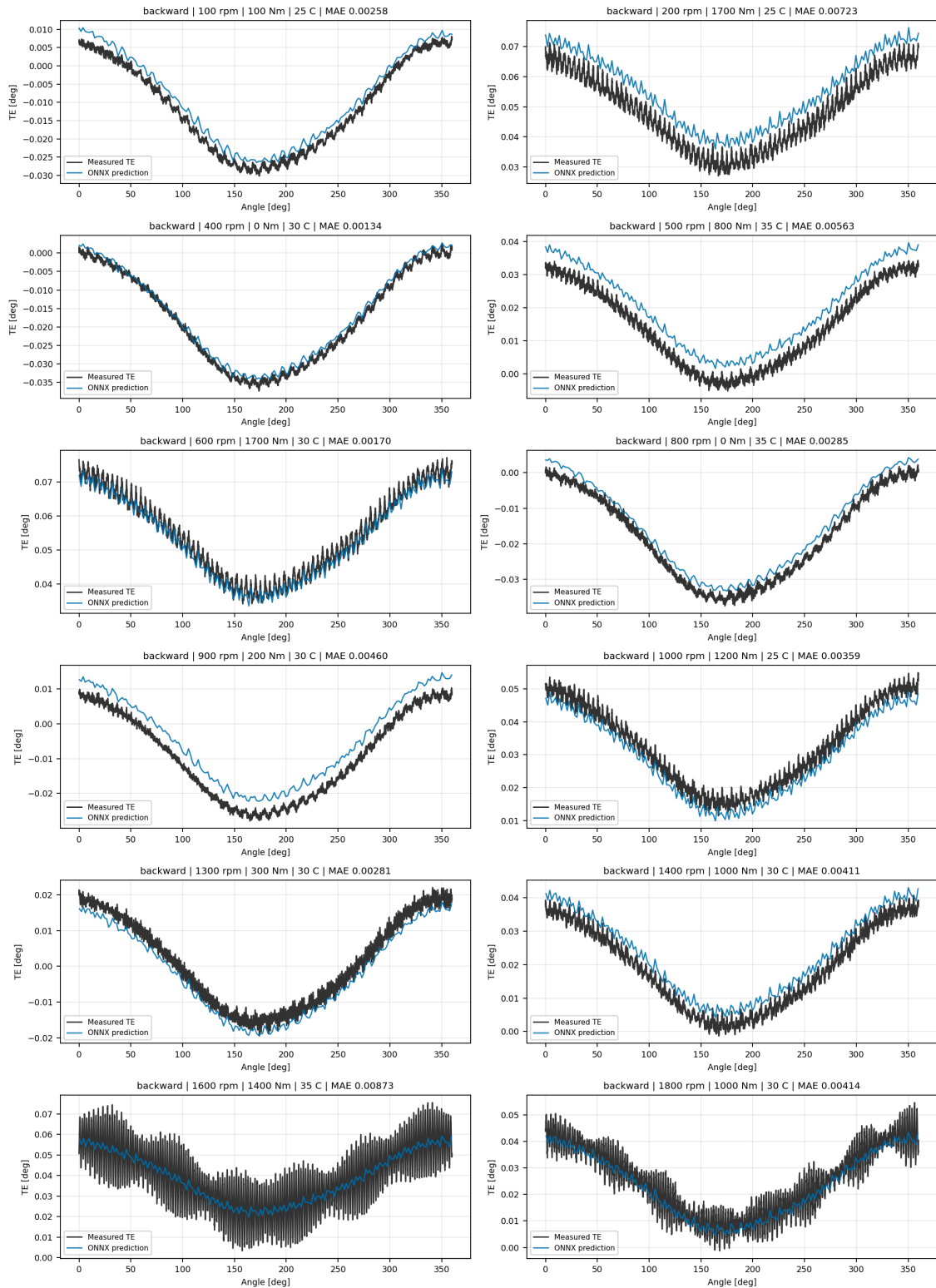
Simplified Dataset + Setpoints | wave4_1_log_cosh_robust_loss | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

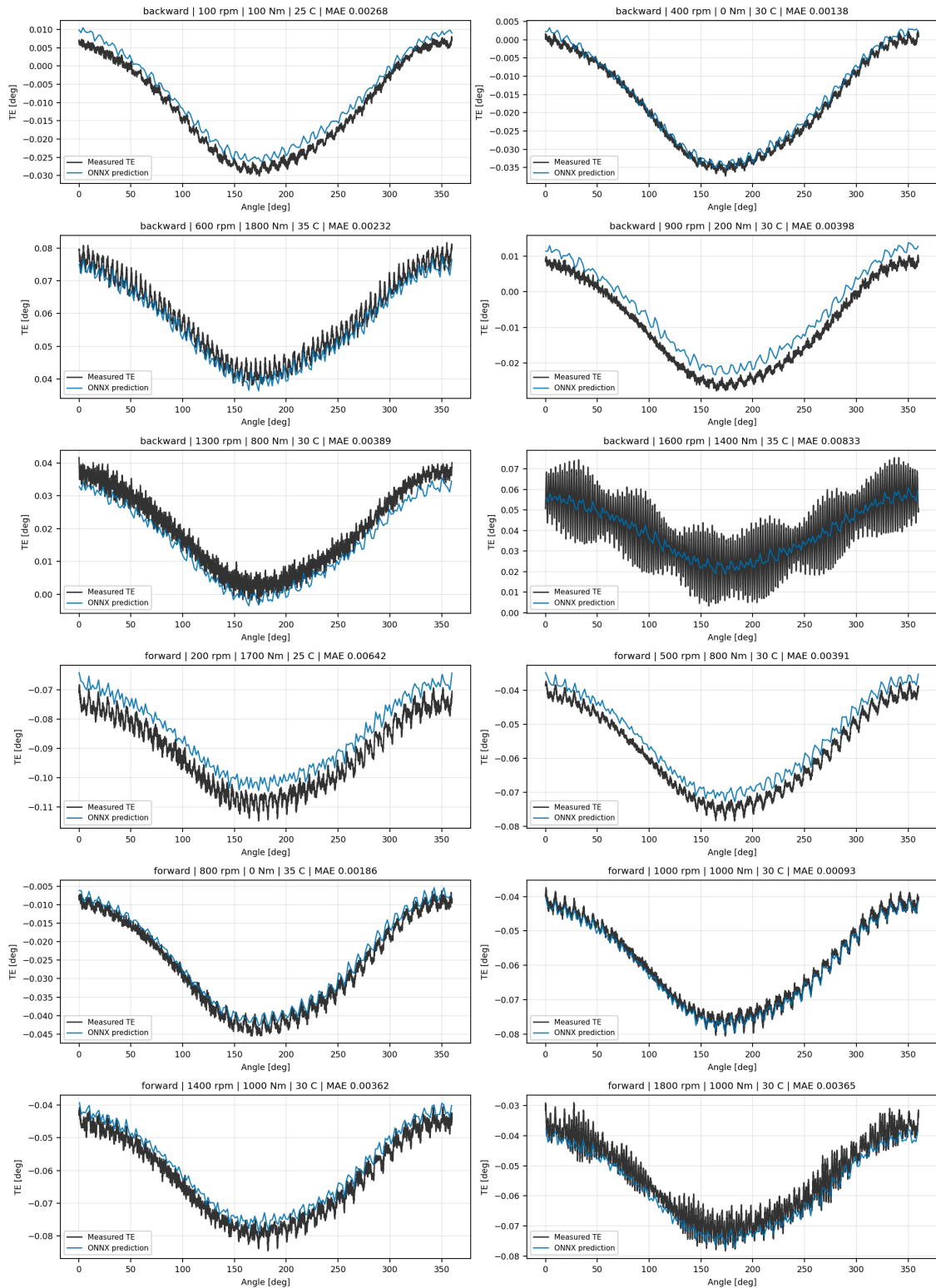
Simplified Dataset + Setpoints | wave_1_log_cosh_robust_loss | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_1_log_cosh_robust_loss | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_1_log_cosh_robust_loss ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_1_log_cosh_robust_loss_fw_polished_setpoints	2026-07-13-07-40-29_te_wave4_1_log_cosh_robust_loss_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_1_log_cosh_robust_loss_bw_polished_setpoints	2026-07-13-08-11-55_te_wave4_1_log_cosh_robust_loss_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_1_log_cosh_robust_loss_global_polished_setpoints	2026-07-13-05-22-10_te_wave4_1_log_cosh_robust_loss_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/forward/2026-07-13-07-40-29_te_wave4_1_log_cosh_robust_loss_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/forward/2026-07-13-07-40-29_te_wave4_1_log_cosh_robust_loss_fw_polished_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=089-val_mae=0.00196005.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/backward/2026-07-13-08-11-55_te_wave4_1_log_cosh_robust_loss_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/backward/2026-07-13-08-11-55_te_wave4_1_log_cosh_robust_loss_bw_polished_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=044-val_mae=0.00196637.ckpt
global	models/polished_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/global/2026-07-13-05-22-10_te_wave4_1_log_cosh_robust_loss_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_log_cosh_robust_loss/global/2026-07-13-05-22-10_te_wave4_1_log_cosh_robust_loss_global_polished_setpoints/python/curve_aware_harmonic_residual_of_offset_probe-epoch=155-val_mae=0.00191211.ckpt

Aggregate Metrics

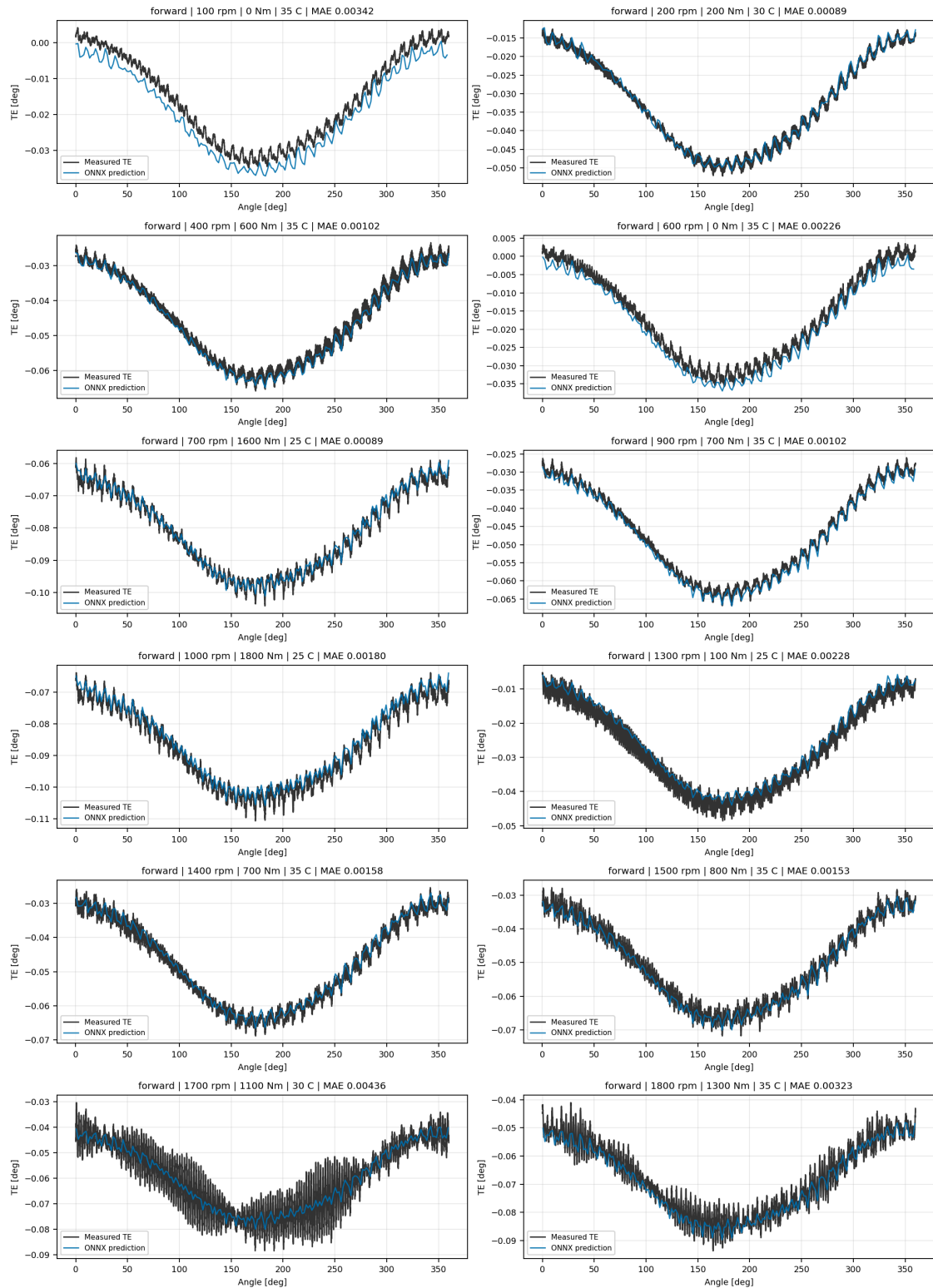
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002048	0.002420	4.519	11.168
backward	94	0.002577	0.003039	4.750	10.910
global	194	0.002200	0.002596	4.382	10.716

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000164	0.001061	0.001600	0.003512
backward	0.000468	0.001011	0.002191	0.005570
global	0.000191	0.000969	0.001804	0.004798

Forward 12-Curve Page

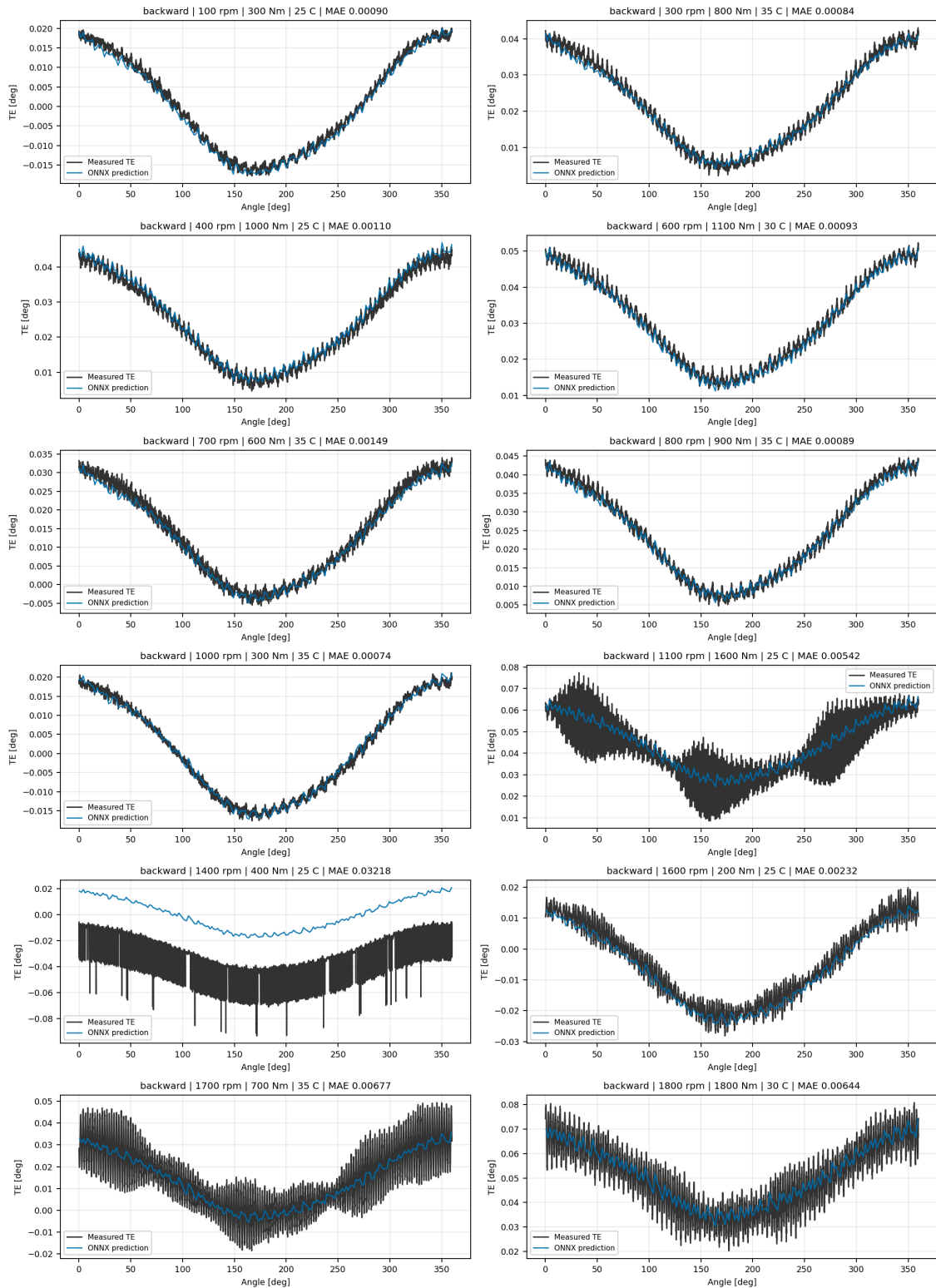
Polished Dataset + Setpoints | wave4_1_log_cosh_robust_loss | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

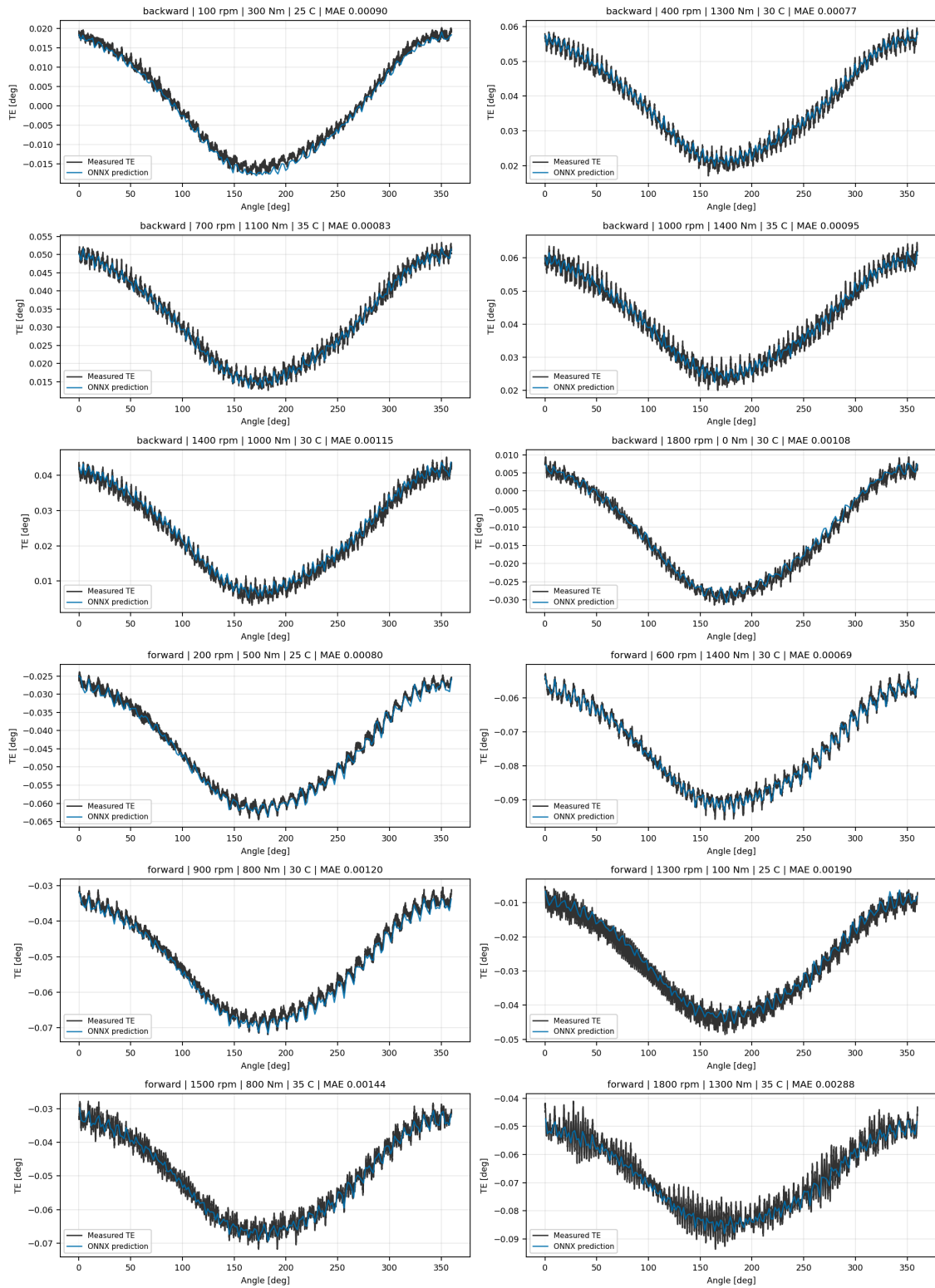
Polished Dataset + Setpoints | wave4_1_log_cosh_robust_loss | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_1_log_cosh_robust_loss | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_1_log_cosh_robust_loss ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_1_log_cosh_robust_loss_fw__polished_actual_values	2026-07-13-09-34-38__te_wave4_1_log_cosh_robust_loss_fw__polished_actual_values	polished_point_v1
backward	te_wave4_1_log_cosh_robust_loss_bw__polished_actual_values	2026-07-13-10-23-44__te_wave4_1_log_cosh_robust_loss_bw__polished_actual_values	polished_point_v1
global	te_wave4_1_log_cosh_robust_loss_global__polished_actual_values	2026-07-13-09-01-50__te_wave4_1_log_cosh_robust_loss_global__polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_1_log_cosh_robust_loss/forward/2026-07-13-09-34-38__te_wave4_1_log_cosh_robust_loss_fw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_log_cosh_robust_loss/forward/2026-07-13-09-34-38__te_wave4_1_log_cosh_robust_loss_fw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=193-val_mae=0.00182691.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_1_log_cosh_robust_loss/backward/2026-07-13-10-23-44__te_wave4_1_log_cosh_robust_loss_bw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_log_cosh_robust_loss/backward/2026-07-13-10-23-44__te_wave4_1_log_cosh_robust_loss_bw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=127-val_mae=0.00187102.ckpt
global	models/polished_dataset/actual_values/exported/wave4_1_log_cosh_robust_loss/global/2026-07-13-09-01-50__te_wave4_1_log_cosh_robust_loss_global__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_log_cosh_robust_loss/global/2026-07-13-09-01-50__te_wave4_1_log_cosh_robust_loss_global__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=110-val_mae=0.00189902.ckpt

Aggregate Metrics

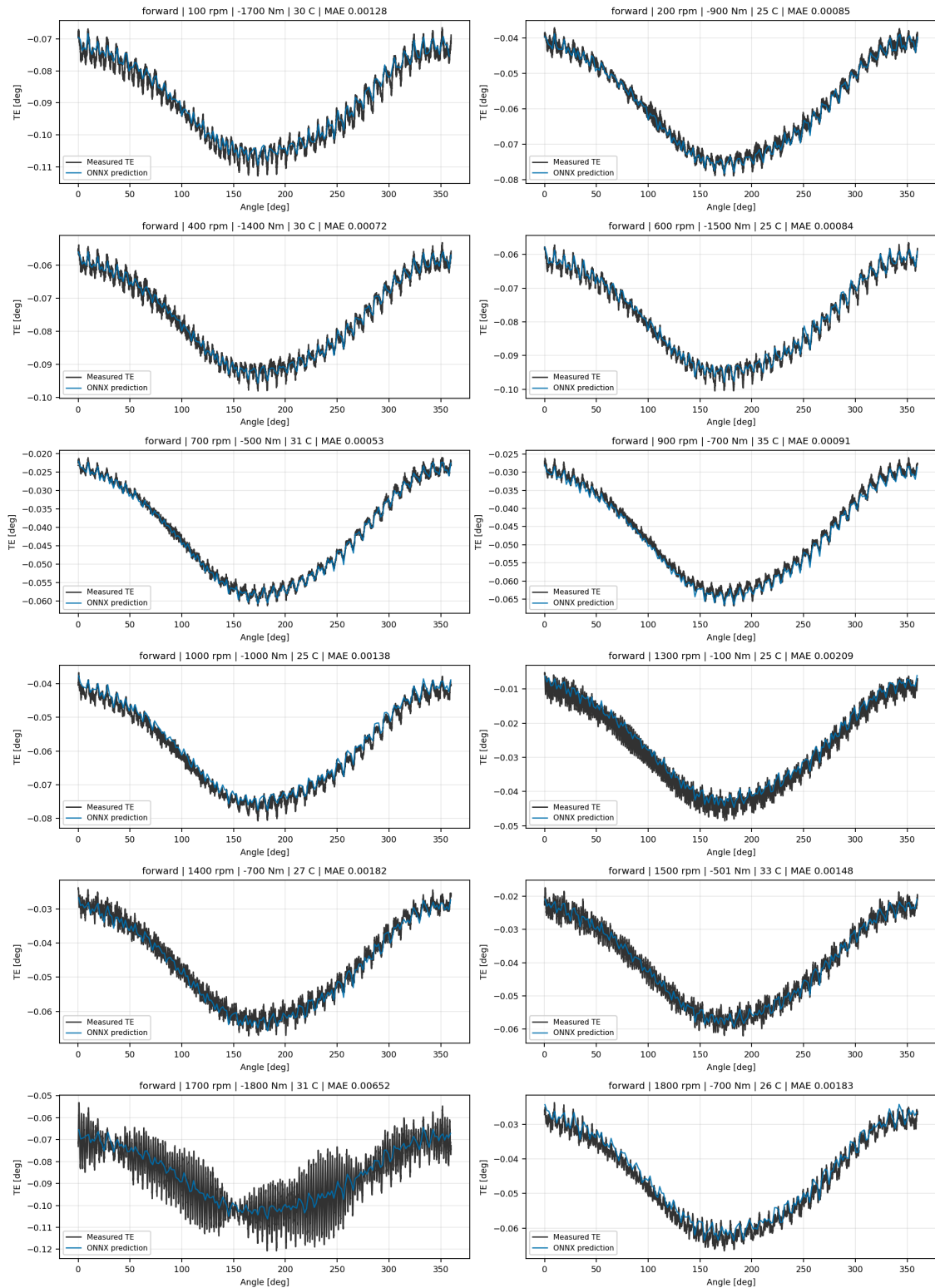
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001749	0.002110	3.797	9.128
backward	94	0.002218	0.002673	4.192	10.475
global	194	0.002046	0.002447	4.160	10.412

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000100	0.000726	0.001551	0.003920
backward	0.000205	0.000642	0.002107	0.005396
global	0.000075	0.000708	0.001818	0.004621

Forward 12-Curve Page

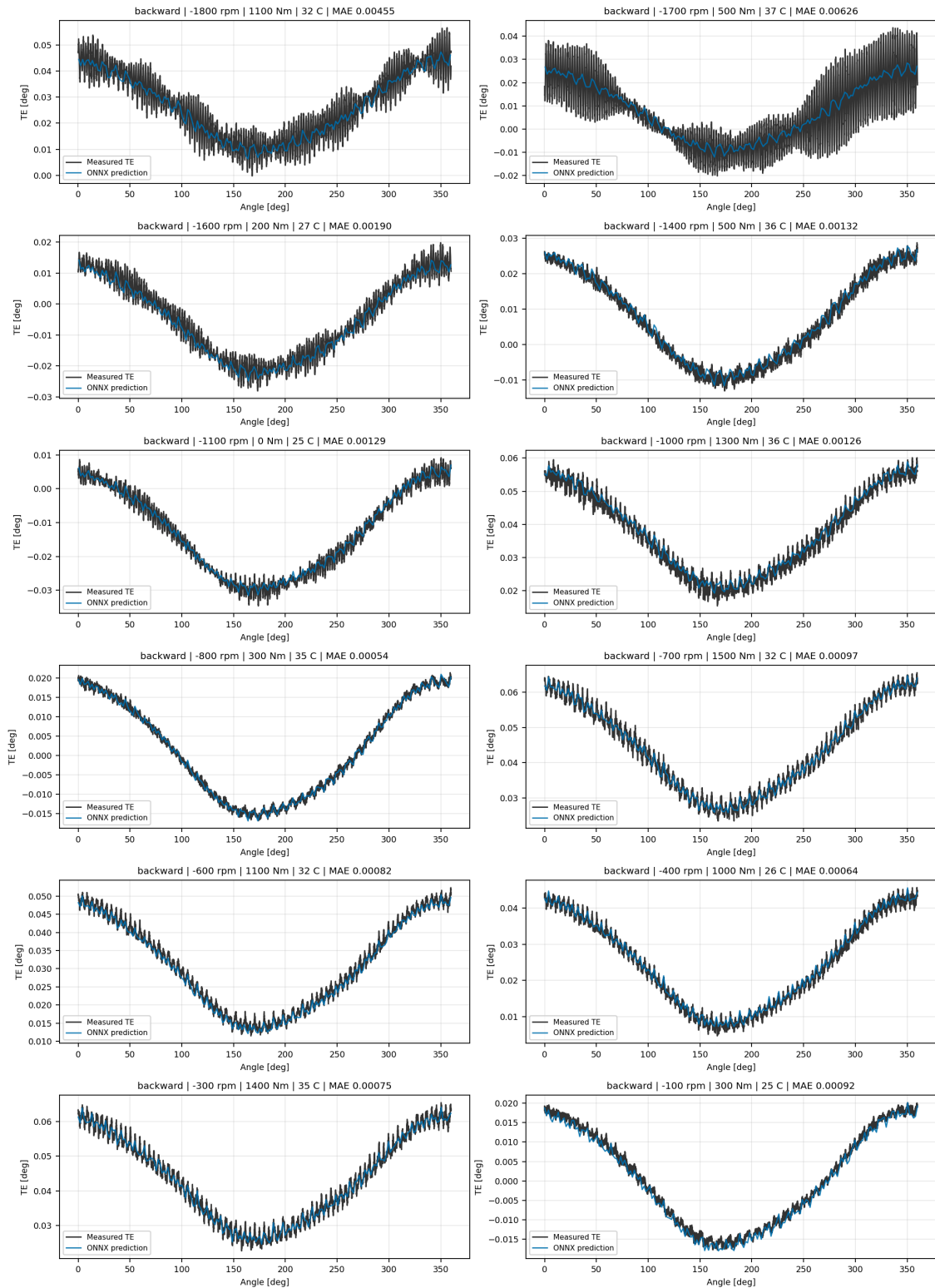
Polished Dataset + Actual Values | wave4_1_log_cosh_robust_loss | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

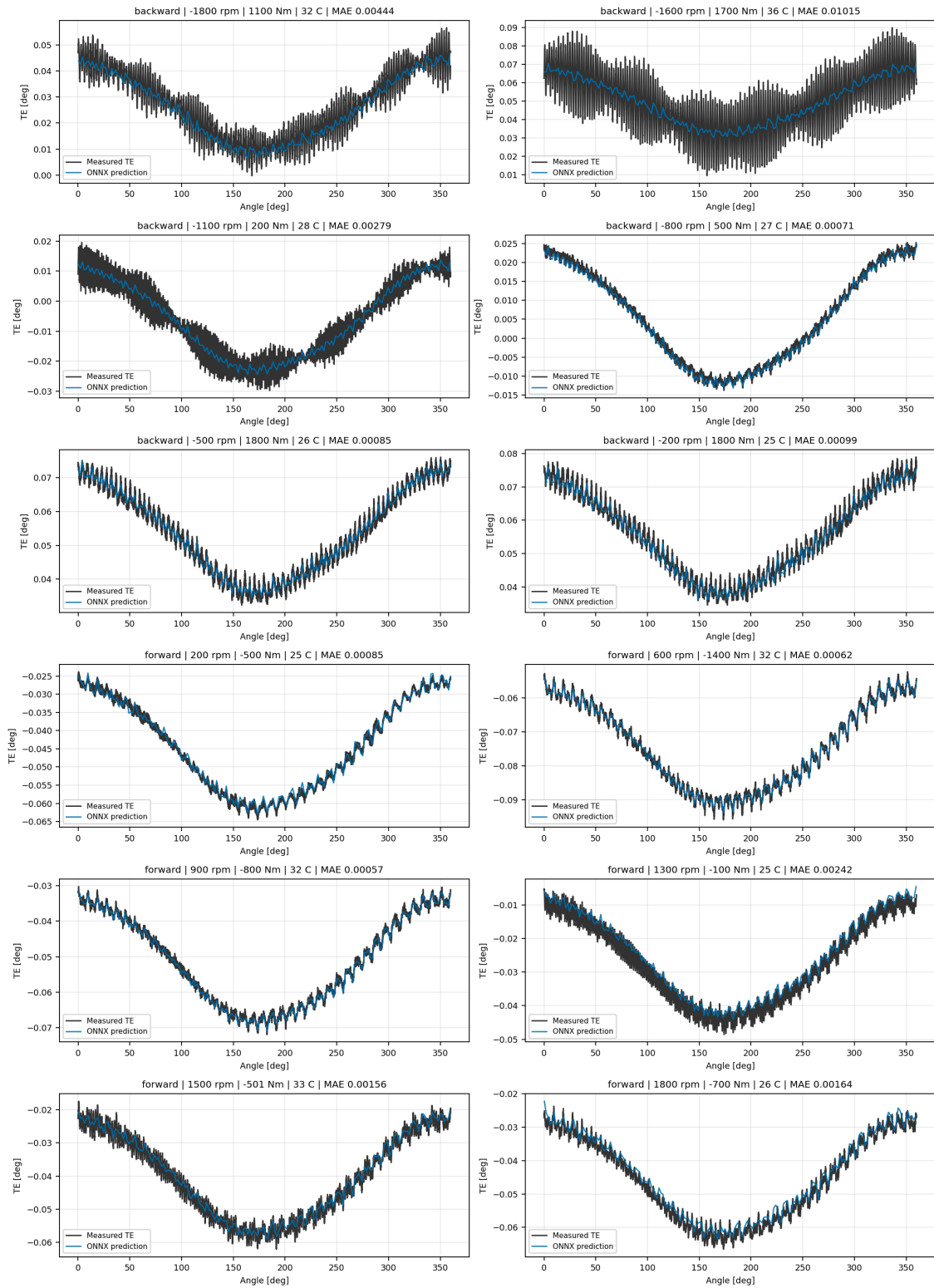
Polished Dataset + Actual Values | wave4_1_log_cosh_robust_loss | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_1_log_cosh_robust_loss | global



polished_dataset_actual_values global 12-curve collage